

DSC 335, Handout 16: Poisson Regression (*Chapter 12*)

LEARNING OBJECTIVES.

- *Build a Bayesian regression model and simulate its posterior.*

Poisson Regression Model

Data: $Y_i | \beta_0, \beta_1, \dots, \beta_p \stackrel{\text{ind.}}{\sim} \text{Pois}(\lambda_i)$ with $\log(\lambda_i) = \beta_0 + \beta_1 X_{i1} + \dots + \beta_p X_{ip}$

Priors: $\beta_0 \sim N(m_0, s_0^2)$

$\beta_1 \sim N(m_1, s_1^2)$

...

$\beta_p \sim N(m_p, s_p^2)$

Interpreting Coefficients:

Regression Assumptions:

Example 1. Suppose we want to use whether or not a student is an athlete and their college GPA to predict how many times they work out in a week.

- (a) Why might a Poisson regression model fit this data better than a Normal regression model?
- (b) Determine a set of priors and fit the model. Give the equation of the regression line based on the posterior median parameter estimates.
- (c) Interpret the coefficients of the model.
- (d) Perform a posterior predictive check. Does our model do a good job at simulating the data structure?
- (e) Create a histogram of predictions for a student-athlete with a GPA of 3.8.